DC KHUSHI

DATA SCIENCE

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OBJECTIVE

Final-year Integrated M.Sc. Data Science student with hands-on experience in machine learning, computer vision, NLP, and data analytics. Skilled in developing end-to-end AI solutions using Python, PyTorch, TensorFlow, and SQL, with research exposure at NIT Tiruchirappalli. Motivated to apply technical skills, creativity, and data-driven thinking to solve real-world problems and focused on building intelligent, meaningful and impactful real-world solutions in industry settings.

EXPERIENCE

Summer Research Intern — May 2025 – Aug 2025

NIT Tiruchirappalli — TRICHY

Project: Intelligent Pothole and Waterlogging Detection System

- Developed a YOLOv8-based computer vision system achieving real-time detection accuracy above baseline models.
- Technologies: YOLOv8, Python, OpenCV, Edge Detection, Deep Learning.
- Deployed and tested models on real-world road imagery datasets

PROJECTS

- **Decision-Making Drone Simulator using Reinforcement Learning** — Jul 2025 – Oct 2025
  Engineered an autonomous drone navigation system using Deep Reinforcement Learning algorithms to enable intelligent decision-making in simulated environments.
  -Technologies: Python, OpenAI Gym, PyTorch, RL Algorithms
- **Crowdsourced Civic Issue Reporting System|SIH 2025 (Phase 2)** — Aug 2024 – Sep 2024
  AI-driven civic platform enabling image/text-based issue reporting with auto geo-tagging, intelligent department routing, and real-time administrative dashboard.
  -Technologies:React.js, Node.js, Python, TensorFlow, OpenCV, MongoDB, Firebase, AWS/GCP
- **Online Meeting Enhancement Suite | CIT 24-Hour Hackathon** — Sep 2025
  Created a comprehensive browser extension and web application to optimize virtual meeting experiences with features like automated note-taking and productivity tools.
  -Technologies: JavaScript, React, Chrome Extension API, Python Backend.
- **Agentic AIOps Predictive Maintenance System** — Jan 2026
  Built a predictive maintenance solution that forecasts machine failure, detects data drift, and provides autonomous, explainable maintenance decisions through an interactive dashboard.
  -Technologies: Python, PyTorch, Transformers, Streamlit, NumPy, SciPy, AIOps

Ongoing Projects:

- **Mockview AI: Automated Mock Interview and Feedback System**
  Currently developing an AI system that analyzes resumes and job descriptions to conduct voice-based interviews and generate automated performance feedback.
- **Trust-Aware Multitask NLP System for Emotion, Hate Speech, and Violence Detection Using LIME XAI** — *RESEARCH BASED*
  Developing a trust-aware, explainable multitask NLP model for emotion, hate speech, and violence detection using LIME-based word-level explanations.
- **Continual Multimodal Retrieval with Lexical Simplification: A VLM-Based Approach for Cross-Domain** — *RESEARCH BASED*
  Working on a vision–language model–based system for cross-domain multimodal retrieval with improved adaptability over time.
- **Agentic AI Smart Meal Planner with Lifestyle-Aware Recommendation Engine**
  Developing an intelligent agent-based system that personalizes meal planning by reasoning over user preferences, available ingredients, and professional work schedules.

EXTRA CURRICULAR ACTIVITIES

- Anokha 2024- Eventide Co-head
- Youth Speaker & Governing Council Member | International Board of Trainers & Coaches (IBTC)
- Central Sports Coordinator | Amrita Vishwa Vidyapeetham | 2025-26
- Public speaking
- Dancing/Drama

EDUCATION

Int.MSc Data Science
Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu
Aug 2022 – Jun 2027 | CGPA: 7.92 / 10

Higher Secondary Education
LK Singhania Education Center, Rajasthan
2021 – 2022 | CBSE | 84.3%
Subjects: BioMaths / English, Maths, Chemistry, Physics, Biology.

Secondary Education
LK Singhania Education Center, Rajasthan
2020 | CBSE | 91%

TECHNICAL INTEREST & SKILLS:

- **Programming**: Python, Java (Basic), C/C++ (Basic)
- **Machine Learning & AI**: Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, NLP, YOLOv8, Transformers, XAI (LIME)
- **Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, Scikit-learn, Pandas, NumPy
- **Databases**: MySQL, PostgreSQL, MongoDB, Cassandra
- **Tools & Platforms**: Git, VS Code, Kaggle, Google Colab, Hadoop, Streamlit, UiPath
- **Web Technologies**: HTML, CSS, JavaScript, React.js, Node.js

CERTIFICATIONS:

- NPTEL certification in Python for Data Science(2024)
- Great Learning Academy– SQL for Data Science

ACHIEVEMENTS & HONOR

- NCC cadet participated the Republic Day Parade 2018-19
- Secured 3rd in Lady Anusuiya Singhania Memorial All India Turn-Coat Hindi Debate - 2021-22
- Awarded the Chairperson's Award for Best Outgoing Student (Batch 2021–22), LKSEC.

SPORTS

- Amrita University Basketball Team Captain, winner of five gold medals(2022-2024), and participant in the South Zone Championship in Nov 2024.
- National Skater(2016)